FINAL YEAR PROJECT

A laboratory proposal. *With the evidence in view.*



Concept render of the assumed 10 × 7 m room. Not a photograph of an installed laboratory.

One 70 m² room, designed as a physical space and a digital service together.

01 A complete kit must share one usable interval.

02 Evidence needs its method, source and settings.

03 Recovery needs a receipt and a responsible owner.

THE STUDENT JOURNEY, SEVEN STEPS

01
Prepare
Read the question, the kit and the safety check.

02
Reserve
One booking holds every item for the same time.

03
Arrive
Check in at the station and confirm the kit.

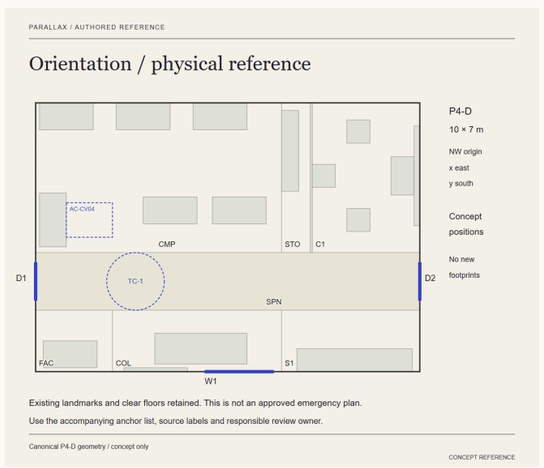
04
Bench
Record each run with its settings.

05
Notebook
Write a claim the evidence supports.

06
Submit
Freeze the revision and wait for the receipt.

07
Return
Count the kit back and close the session.

THE ROOM



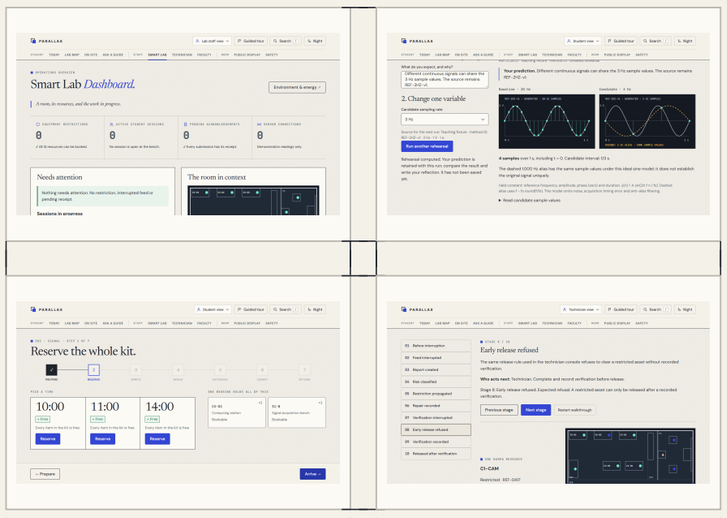
Seven zones total 70 m²: computing 24.96, capture 10.92, circulation 15.00, collaboration 7.04, signal work 5.76, faculty 3.20, storage 3.12. A 1.5 m spine and a 1.2 × 0.9 m accessible approach are kept clear in the model.

SMART FEATURES, AND HOW REAL EACH ONE IS

Digital experiment manual	WORKING
Equipment lookup by label	WORKING
AI assistant	WORKING
Digital twin	WORKING
Equipment health	WORKING
IoT sensors	SIMULATED
Smart energy monitoring	SIMULATED
AR / VR learning	PARTLY
Voice assistant	CONCEPT

Working means it runs in the local prototype with example records, not that an institutional service is connected.

THE PROTOTYPE



A working local prototype: student journey, smart dashboard, fault recovery, lab map, on-site lookup and a local AI guide that cites its source. Open the Guided tour in the top bar to follow the demonstration order.

ROOM
70 m²
An assumed 10 × 7 m envelope, 3 m high, divided into seven zones.

BOOKABLE RESOURCES
13
Stations, the capture area, camera, lights, tripod, target and signal bench.

REVIEWED SOURCES
14
Secondary research only. Zero local participants.

WIREFRAMES
15
Annotated boards from preparation to technician recovery.

DESIGN SYSTEM
40
Component families, with 87 variants in the Figma library.

AUTOMATED TESTS
155
139 application tests and 16 assistant-service tests pass.

Honest boundaries. Research is secondary only, with fourteen reviewed sources and no local participants. Sensor readings, energy figures and bookings are examples. The room is a design model, not a surveyed space. The AI guide is a real local model limited to the project's reviewed guides.